Hints

Voltkart · Session 1 — one nudge per question

How to use these

> **Tip** Use a hint only after you've genuinely tried the question. Each hint points you at the right tables and concepts — it never hands you the query.

Hints by question

Question 1 — Use `fact_orders`, `dim_customer`, `dim_employee`. Concepts: INNER JOINs, WHERE on status, TOP, ORDER BY.

Question 2 — Use `dim_customer`, `fact_orders`. Concepts: anti-join with `NOT EXISTS` (think about why `NOT EXISTS` is safer and usually faster than `NOT IN`).

Question 3 — Use `fact_order_items`, `fact_orders`, `dim_product`, `dim_category`. Concepts: SUM of `line_amount`, `RANK() OVER (PARTITION BY category ORDER BY revenue DESC)`, then filter to rank ≤ 3.

Question 4 — Use `fact_orders`. Concepts: a month bucket via `CONVERT(char(7), order_date, 126)`; `SUM() OVER (...)` for the running total; `LAG()` for the month-over-month change.

Question 5 — Use `fact_orders`. Concepts: a per-customer SUM, then `NTILE(4) OVER (ORDER BY spend)`, then GROUP BY the quartile.

Question 6 — Use `dim_category`. Concepts: a recursive CTE — anchor at 'Computers', recurse on `parent_category_id`; track a depth counter and build the path with string concatenation (CAST to NVARCHAR).

Question 7 — Use `dim_employee`, `fact_orders`. Concepts: a recursive CTE that enumerates each employee's subtree (pair every root with all of its descendants, including itself), then attribute rep orders and SUM up the tree.

Question 8 — Use `fact_orders` (target) and `stg_orders_incr` (source). Concepts: `MERGE … ON order_id`, `WHEN MATCHED THEN UPDATE`, `WHEN NOT MATCHED BY TARGET THEN INSERT`. End the statement with a semicolon.

Question 9 — Use `dim_product` (target) and `cdc_product_changes` (source). Concepts: a single MERGE with `WHEN MATCHED AND operation='U'` → UPDATE, `WHEN MATCHED AND operation='D'` → DELETE, `WHEN NOT MATCHED AND operation='I'` → INSERT.

Question 10 — Use `fact_orders`, `fact_order_items`. Concepts: SARGability (avoid `YEAR()` on the column — use a date range), replace the correlated subquery with a pre-aggregated

CTE or join, and add a covering index with INCLUDE. Read the logical reads from `SET STATISTICS IO`.

Question 11 — Use `fact_orders`, `dim_customer`. Concepts: gaps-and-islands — turn each active month into an integer, subtract `ROW_NUMBER()` per customer to label consecutive runs, then take the longest run.